Potato Diseases

Identification, causes, favourable conditions, management and verified remedies

Ten diseases of potato (Solanum tuberosum), India focus, including early blight and late blight

Primary sources: ICAR-Central Potato Research Institute (CPRI), TNAU Agritech Portal, CIB&RC (Insecticides Act, 1968) Major Uses of Pesticides up to 30.09.2025, and ICAR-linked research. Compiled 21 October 2026.

Contents

About this document

Summary table

1. Late Blight
2. Early Blight (Alternaria Leaf Spot)
3. Black Scurf and Stem Canker
4. Common Scab
5. Bacterial Wilt (Brown Rot)
6. Soft Rot and Blackleg
7. Fusarium Dry Rot
8. Silver Scurf
9. Potato Viruses (Leaf Roll, Y, X and Apical Leaf Curl)
10. Potato Wart (Quarantine Disease)

Appendix A. Cross-check log

Appendix B. Image-classification guide for the model classes

About this document

Scope and method

Each disease chapter follows the same six headings: identification; causes and progression; favourable conditions; prevention and management; remedies and treatment; and sources. Disease descriptions come mainly from ICAR-Central Potato Research Institute (CPRI) and TNAU material. Chemical recommendations are included only where they could be matched to the CIB&RC list of registered potato uses or to a published ICAR-CPRI or TNAU recommendation, and each is tagged to show how it was verified.

Verification tags used in the chemical tables

V1 - Verified against CIB&RC. The product is listed for POTATO and this disease (or a named group including it) in the CIB&RC 'Major Uses of Pesticides - Fungicides' compilation, up to 30.09.2025. Doses are per hectare as printed.

V2 - ICAR-CPRI or TNAU recommendation. A matching potato label claim was NOT found in the CIB&RC list in this review; confirm current registration and the product label.

-- - No chemical option verified for this disease; none is recommended.

Important cautions

- The CIB&RC compilation used for V1 is 'based on certificate issued' up to 30.09.2025 and carries a disclaimer that it is for guidance and not for legal purposes. A hosted copy was read, not the live register. Registrations and labels change; check the current label and the CIB&RC register (ppqs.gov.in) and follow the pre-harvest interval (PHI) printed on the label.
- Doses are per hectare as printed in CIB&RC; TNAU and trial advice are given as concentrations. Some printed values look odd or lack units (for example a likely misprint in the chlorothalonil + mancozeb row, PHIs printed as 'at the end of harvest' or '3-5', and seed treatment units). These are flagged where they appear; the label governs.
- Rotate fungicide groups across sprays, begin protection before disease appears for late blight, use PPE, follow label safety instructions and keep products away from water bodies.
- The ICAR-CPRI article (CP1) is the main source for disease descriptions and non-chemical measures; TNAU pages and an ICAR-linked late blight trial supplement it. No MPKV Rahuri, PDKV Akola or UC IPM potato page was retrieved, so none is cited. Sources marked supplementary support only the points where they are cited.

Summary table

Disease	Pathogen / type	Main non-chemical measures	Chemical status
Late Blight	Phytophthora infestans (oomycete; tuber- and soil-borne)	Certified seed, resistant varieties, rotation, culling, forecast-based sprays	Many V1 options (50 CIB&RC rows)
Early Blight	Alternaria solani complex (fungal; residue-borne)	Sanitation, volunteer and weed control, balanced nutrition (K, Zn)	Many V1 options (23 CIB&RC rows)
Black Scurf and Stem Canker	Rhizoctonia solani AG 3-PT (seed- and soil-borne)	Clean seed, rotation, timely harvest	V1 seed tuber treatments (9 rows)
Common Scab	Streptomyces spp. (seed- and soil-borne)	Soil moisture after tuber initiation, pH, rotation, green manure	V1 thiram seed treatment
Bacterial Wilt (Brown Rot)	Ralstonia solanacearum (soil- and tuber-borne)	Clean seed, clean fields, stable bleaching powder	V1 streptocycline; V2 bleaching powder
Soft Rot and Blackleg	Pectobacterium / Dickeya spp.	Avoid injury, ventilation, clean seed	V1 streptocycline; V2 boric acid
Fusarium Dry Rot	Fusarium spp. (soil- and debris-borne)	Minimise wounds, potassium, storage care	None V1; V2 boric acid dip
Silver Scurf	Helminthosporium solani (tuber-borne)	Clean seed, dry stores	None V1; V2 boric acid dip
Potato Viruses	PLRV, PVY, PVX, PALCV (aphid-borne)	Virus-free seed, vector management, rogueing	None; fungicides do not apply
Potato Wart	Synchytrium endobioticum (quarantine)	Quarantine and clean seed	None verified

Quick separation of look-alikes: dry, papery, angular dark-brown spots with concentric rings on lower leaves point to early blight; water-soaked pale green lesions that turn brown or purplish-black, with white mildew under the leaf in humid weather, point to late blight; rough corky pits on tuber skin point to common scab; black crusty sclerotia on tubers with girdled stems point to black scurf; wilting with white streaming from cut stems in water points to bacterial wilt; slimy, foul-smelling tuber rot and a black stem base point to soft rot and blackleg; dry shrunken wrinkled tubers point to Fusarium dry rot; and leaf rolling or mosaic points to viruses.

1. Late Blight

Potato | Phytophthora infestans | Oomycete (fungus-like; tuber- and soil-borne; wind-dispersed sporangia)

At a glance. The most destructive potato disease in India: water-soaked leaf lesions with white mildew beneath that can destroy a crop within a week in cool, humid weather. ICAR-CPRI estimates average annual losses of about 15% in India. Certified seed, resistant varieties, forecast-based or schedule-based protectant sprays and a rotation of fungicide groups are the core of management.

1. Identification

Symptoms

- Crop / stage: leaves, stems, petioles and tubers at any growth stage; sepals, flowers and berries can also be infected (CP1).
- Leaves: water-soaked, irregular, pale green lesions, mostly near the tip and margins, that grow rapidly into large brown to purplish-black necrotic spots. A white mildew (sporangiophores and sporangia) shows on the lower leaf surface around the edges of lesions in humid weather; in dry weather the water-soaked areas dry up and turn brown (CP1, TP1).
- Stems and petioles: light to dark brown lesions encircle the stem, which weakens and collapses at that point; the stem often breaks at the nodes and the plant topples (CP1, TP1).
- Tubers: irregular reddish-brown to purplish areas that extend into the tuber; cut tubers show rusty-brown necrosis spreading from the surface to the centre. Infected tubers are usually hard, dry and firm but can be invaded by soft-rot bacteria in the field and store (CP1, TP1).
- Under favourable conditions the whole crop looks blighted and can be destroyed within a week (CP1).

Distinguishing features

- Early blight lesions are dry, papery and angular, often with concentric rings, and carry no white sporulation; late blight lesions are water-soaked with white mildew beneath (CP1, AHDB).
- Alternaria spores cannot be seen on the leaf even with a hand lens, and there is no milky ring of sporulation round the lesion underside (AHDB).

2. Causes and progression

- Pathogen: the oomycete Phytophthora infestans; sporangia are lemon-shaped and release biflagellate zoospores (CP1, DEY, TP3).
- Survival: infected tubers and infected soil are the primary source; diseased tubers carry the disease from crop to crop (TP1). Oospores can survive up to three years in host debris in soil (CP1).
- Spread: airborne sporangia from infected foliage; tubers in the soil are infected by rain- or dew-borne sporangia trickling down from diseased foliage (CP1, TP1).
- Progression: cell death occurs within 2 to 4 days under favourable weather and the entire crop can collapse (DEY). Pathogen populations change over time, so resistant varieties and fungicide performance can be affected (CP1, DEY).
- Losses: about 15% average annual loss in India, equal to about 8.0 million tonnes (CP1); yield losses of 11 to 74% depending on climate are cited (DEY).

3. Favourable conditions

- Moderate temperature (15 to 20 C) with relative humidity above 80% (CP1).
- Cool (15 to 18 C), moist weather with RH 80% or more in north-eastern India (DEY).
- TNAU: cool (12 to 15 C), humid (above 90%) weather with rain alternating with warm (20 C), moist periods (TP3); relative humidity above 90%, 10 to 25 C, night temperature about 10 C, cloudiness before rain and at least 0.1 mm rain (TP1).
- No or short rotation (fields without rotation have earlier infection) and nitrogen-rich foliage, which raises late blight severity (CP1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use certified seed tubers from a reliable source and do not cut seed tubers; the knife spreads pathogens and cut surfaces pick up disease from soil (CP1).
- Grow resistant varieties. TNAU lists Kufri Naveen, Kufri Jeevan, Kufri Alankar (spelt Alenkar by TNAU) and Kufri Moti (TP3). Resistance can break down because the pathogen is adaptive; Kufri Jyoti was the susceptible check in a Tripura trial (DEY).

- Rotate: at least three years between potato crops greatly reduces early infection from soil inoculum (CP1).
- Clear weeds and crop remnants and burn or deep-bury them; deep ploughing twice, 20 to 30 days apart, in peak summer (plains) or winter (hills) exposes inoculum to above 40 C or below 5 C (CP1).
- Avoid injury at harvest and discard visibly infected tubers before storage to limit tuber contamination (TP1).
- Adequate potassium reduces several diseases including late blight, and potassium phosphate or phosphorous acid suppresses late blight alone or with fungicides (CP1).
- Use forecasts: JHULSACAST (western Uttar Pradesh, Punjab, Terai of Uttarakhand, plains of West Bengal) and the pan-India INDO-BLIGHTCAST (since 2016) support need-based sprays (CP1).

Regional note

No Maharashtra-specific late blight recommendation was found in the sources reviewed. ICAR-CPRI guidance applies; regional forecast models exist for other regions (CP1).

5. Remedies and treatment

Biological and non-registered options

- Trichoderma viride, T. harzianum, Pseudomonas fluorescens, P. aeruginosa and Bacillus subtilis have been effective against several potato pathogens in experiments (CP1). The CIB&RC bio-fungicide list has no entry for potato (CIBB), so none is recommended for late blight here.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Amisulbrom 20% SC	Late blight	500 ml/ha (100 g a.i.)	375-500 L	V1	CIB&RC; PHI 19 d
Azoxystrobin 23% SC	Late blight	500 ml/ha (125 g a.i.)	500 L	V1	CIB&RC; PHI 12 d
Captan 50% WG	Early and late blight	1500 g/ha	500 L	V1	CIB&RC; PHI 21 d
Captan 50% WP	Early and late blight	2.5 kg/ha	750-1000 L	V1	CIB&RC; PHI not stated
Captan 75% WP	Early and late blight	1667 g/ha	1000 L	V1	CIB&RC; PHI 8 d
Chlorothalonil 75% WP	Early and late blight	0.875-1.25 kg/ha	600-800 L	V1	CIB&RC; PHI 14 d
Copper oxychloride 50% WP	Early and late blight	2.5 kg/ha	750-1000 L	V1	CIB&RC; PHI not stated
Copper hydroxide 53.8% DF	Late blight	1500 g/ha	500 L	V1	CIB&RC; PHI 22 d
Copper sulphate 2.62% SC	Early and late blight	1 L/ha	500 L	V1	CIB&RC; PHI 3 d
Cyazofamid 34.5% SC	Late blight	200 ml/ha (80 g a.i.)	500 L	V1	CIB&RC; PHI 27 d
Dimethomorph 50% WP	Late blight	1000 g/ha	750 L	V1	CIB&RC; PHI 16 d
Ethaboxam 40% SC	Late blight	500-663 ml/ha	500 L	V1	CIB&RC; PHI 11 d
Hexaconazole 2% SC	Early and late blight	3.0 L/ha (60 g a.i.)	500 L	V1	CIB&RC; PHI 21 d
Kresoxim-methyl 44.3% SC	Early and late blight	500 ml/ha (250 g a.i.)	500 L	V1	CIB&RC; PHI 25 d
Mancozeb 35% SC	Early and late blight	0.5% (500 g/100 L)	500 L or as required	V1	CIB&RC; PHI 35 d
Mancozeb 75% WG	Late blight	1000 g/ha	500 L	V1	CIB&RC; PHI 3-5 d as printed
Mancozeb 75% WP	Late and early blight	1.5-2 kg/ha (TNAU: 0.2% protective spray)	750 L	V1	CIB&RC; PHI not stated; TP1
Mandipropamid 23.4% SC	Late blight	0.8 ml/L	500-750 L	V1	CIB&RC; PHI 40 d
Metiram 70% WG	Late and early blight	2000 g/ha	500-750 L	V1	CIB&RC; PHI 17 d
Oxathiapiprolin 10.1% OD	Late blight	200 ml/ha (20 g a.i.)	500 L	V1	CIB&RC; PHI 22 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Picarbutrazox 9.53% SC	Late blight	1000-1250 ml/ha	750-1000 L	V1	CIB&RC; PHI 34 d
Propineb 70% WP	Early and late blight	0.30% (300 g/100 L)	As required	V1	CIB&RC; PHI 15 d
Zineb 75% WP	Early and late blight	1.5-2 kg/ha	750-1000 L	V1	CIB&RC; PHI not stated
Ametoctradin 27% + Dimethomorph 20.27% SC	Late blight	800-1000 ml/ha	500 L	V1	CIB&RC; PHI 32 d
Azoxystrobin 8.3% + Mancozeb 66.7% WG	Early and late blight	1500 g/ha	500 L	V1	CIB&RC; PHI 25 d
Azoxystrobin 11% + Tebuconazole 18.3% SC	Early and late blight	750 ml/ha	500 L	V1	CIB&RC; PHI not printed
Captan 70% + Hexaconazole 5% WP	Early and late blight	500-1000 g/ha	500 L	V1	CIB&RC; PHI 21 d
Carbendazim 12% + Mancozeb 63% WP	Early and late blight, black scurf	1750 g/ha	500 L	V1	CIB&RC; PHI 47 d
Chlorothalonil 12.5% + Mancozeb 62.5% WG	Early and late blight	1750 g/ha	500 L	V1	CIB&RC; PHI 35 d (a.i. printed 2178.75, a likely misprint of 218.75)
Copper oxychloride 47% + Metalaxyl 8% WP	Early and late blight	2127.65 g/ha	700 L	V1	CIB&RC; PHI 34 d
Cyazofamid 6.2% + Mancozeb 60.8% WP	Early and late blight	1250 g/ha	500 L	V1	CIB&RC; PHI 22 d
Cymoxanil 8% + Mancozeb 64% WP	Late blight	1500 g/ha	500-750 L	V1	CIB&RC; PHI 10 d
Dimethomorph 12% + Pyraclostrobin 6.7% WG	Late blight	1250 g/ha	500 L	V1	CIB&RC; PHI 26 d
Famoxadone 16.6% + Cymoxanil 22.1% SC	Late blight	500 (ml) /ha as printed (210 g a.i.)	500 L	V1	CIB&RC; PHI 40 d
Fenamidone 10% + Mancozeb 50% WG	Late blight	1250-1500 g/ha	500 L	V1	CIB&RC; PHI 30 d
Fluopicolide 6.25% + Propamocarb HCl 62.5% SC	Late blight	1.25-1.5 L/ha	500 L	V1	CIB&RC; PHI 19 d
Iprovalicarb 5.5% + Propineb 61.25% WP	Late blight	2.0 kg/ha	500 L	V1	CIB&RC; PHI 26 d
Kresoxim-methyl 15% + Chlorothalonil 56% WG	Early and late blight	1000 g/ha	500 L	V1	CIB&RC; PHI 23 d
Mancozeb 69.7% + Cyazofamid 5.3% WG	Late blight	1500 g/ha	500 L	V1	CIB&RC; PHI 36 d
Metalaxyl-M 4% + Mancozeb 64% WP	Late blight	2500 g/ha (0.25%)	500-1000 L	V1	CIB&RC; PHI 24 d
Metalaxyl-M 3.3% + Chlorothalonil 33.1% SC	Late blight	1000 ml/ha (0.2%)	500 L	V1	CIB&RC; PHI 34 d
Metalaxyl-M 3.9% + Mancozeb 64% WG	Early and late blight	2500 g/ha	500 L	V1	CIB&RC; PHI 32 d
Metalaxyl 8% + Mancozeb 64% WP	Late blight	2500 g/ha (0.25%)	1000 L	V1	CIB&RC; not less than 7 weeks
Metiram 44% + Dimethomorph 9% WG	Late blight	2500 g/ha	500 L	V1	CIB&RC; PHI 11 d
Metiram 55% + Pyraclostrobin 5% WG	Late blight	1500-1750 g/ha	500 L	V1	CIB&RC; PHI 15 d
Oxathiapiprolin 4.61% + Amisulbrom 23.08% SE	Late blight	312.5 (ml)/ha as printed (90 g a.i.)	500 L	V1	CIB&RC; PHI 43 d
Oxathiapiprolin 0.6% + Mancozeb 60% WG	Late blight	2500 g/ha	500 L	V1	CIB&RC; PHI 20 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Propineb 70% + Cymoxanil 6% WP	Early and late blight	2000-2250 g/ha	500 L	V1	CIB&RC; PHI printed as 'at the end of harvest'
Valifenalate 6% + Mancozeb 60% WG	Late blight	2000-2500 g/ha	500 L	V1	CIB&RC; PHI 20 d
Spray schedule: mancozeb 75 WP 2.5 g/L at 35 DAS; cymoxanil + mancozeb 2.5 g/L at 42 DAS; mancozeb at 49 DAS; cymoxanil + mancozeb at 56 DAS	Late blight (best of eight schedules)	As stated, 7-day intervals	-	V2	ICAR / KVK trial, Tripura (DEY); each product is V1 above; check each PHI before harvest

Spray protectant (contact) fungicides before the disease appears, then alternate with systemic or translaminar + contact products. In an ICAR-linked trial, one preventive mancozeb spray at 35 days after sowing followed by alternating cymoxanil + mancozeb and mancozeb sprays gave the lowest disease and highest yield (DEY).

Rotate fungicide groups across sprays and keep to each PHI; the PHIs above are as printed for potato in the CIB&RC list. Where doses are given as a percentage, use the volume the label states.

Doses are per hectare as printed in the CIB&RC list; TNAU and the trial give concentrations. Where dose units were not printed (marked 'as printed'), follow the label.

6. Sources and citations

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[TP1] TNAU Agritech Portal - Potato: Late blight (Phytophthora infestans). https://agritech.tnau.ac.in/crop_protection/crop_prot_crop%20diseases_veg_potato_1.html

[TP3] TNAU Agritech Portal - Post-harvest diseases, vegetables, potato: Late blight. https://agritech.tnau.ac.in/crop_protection/potato_phdiseases_4.html

[DEY] Dey, U. et al. (2024). Management of late blight (Phytophthora infestans) of potato through efficient spray schedule of fungicides in the north-eastern Himalayan region of India. Indian Journal of Agricultural Sciences 94(8): 841-846. https://epubs.icar.org.in/index.php/IJAgS/article/download/147373/55311/424214

[AHDB] AHDB (UK) Knowledge Library: Alternaria (early blight) of potato (supplementary, non-Indian source). https://archive.ahdb.org.uk/knowledge-library/alternaria

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

2. Early Blight (Alternaria Leaf Spot)

Potato | Alternaria solani (with A. alternata and other Alternaria species) | Fungal (residue-borne, wind-dispersed)

At a glance. Dry, angular dark-brown leaf spots with concentric rings (bull's-eye) that spread from lower leaves in warm, humid weather with alternate wet and dry spells. It also dents tuber quality. Sanitation, good nutrition and well-timed protectant sprays are the main tools; several products are labelled for it in India.

1. Identification

Symptoms

- Leaves: small, round, oval or angular, dark brown to black, dry and papery spots with angular margins, usually limited by leaf veins; they may or may not show concentric rings, which are most obvious in large blotches and give a bull's-eye or target look (CP1, TP2).
- Tissue around the spots often turns chlorotic and yellow, suggesting toxins; spots can coalesce and the field then looks severely blighted (CP1).
- Tubers: dark, sunken, circular to irregular, shallow lesions separated from healthy tissue by a purplish-brown dry cork layer (CP1); TNAU notes 'shot holes' in tubers (TP2).
- It is present in both the hills and the plains (TP2).

Distinguishing features

- Late blight lesions are water-soaked and bear white mildew beneath the leaf (CP1).
- Early blight symptoms can be mistaken for magnesium deficiency, manganese toxicity or Verticillium wilt (AHDB).
- Alternaria on potato is a disease complex: A. alternata (brown spot) and A. solani often occur together on one leaf and their symptoms are hard to separate (CP1).

2. Causes and progression

- Pathogen: Alternaria solani is recognised as the main cause; A. alternata is associated with brown spot; eleven Alternaria species have been linked to foliar potato disease worldwide (CP1, AHDB).
- Survival: conidia and mycelium in soil or debris of affected plants can stay viable for more than 17 months and cause primary infection in the next crop (TP2). The disease carries over on volunteer tomato and potato plants and weedy Solanaceae such as nightshade and horsenettle (CP1, AHDB).
- Spread: secondary infection is more important; conidia from primary spots are carried long distances by wind, and to nearby plants by rain and insects (TP2).
- In Israeli work, early blight was described as principally a disease of senescing plants (SHT, supplementary).

3. Favourable conditions

- Warm, humid conditions (24 to 29 C) and alternating wet and dry periods (CP1).
- TNAU: dry, warm weather with intermittent rain, 25 to 30 C, poor crop vigour and poorly manured crops (TP2).
- Crop stress and poor nutrition raise severity (AHDB, TP2).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Remove weeds and volunteer potato and tomato plants; clear crop residue and rotate, since long rotations lower incidence (CP1, AHDB).
- Avoid crop stress and give balanced fertiliser and irrigation (AHDB, TP2). Nitrogen-rich foliage increases resistance to Alternaria but raises late blight severity (CP1).
- Adequate potassium and zinc reduce early blight (CP1).
- Forecasting models exist for early blight (CP1).
- In Israeli work, fungicide programmes applied against late blight in the spring crop were enough to suppress early blight (SHT, supplementary), and AHDB likewise says effective late blight control remains the most important factor in fungicide strategy (AHDB).

Regional note

No Maharashtra-specific early blight recommendation was found in the sources reviewed.

5. Remedies and treatment

Biological and non-registered options

- No biological product has a potato claim in the CIB&RC bio-fungicide list (CIBB). The tomato early blight entries (Pseudomonas fluorescens 1.75% WP, Bacillus subtilis 1.5% L.F.) are for tomato and are not reproduced for potato.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Zineb 75% WP	Early blight	1.5-2 kg/ha (TNAU: 0.2% very early, repeat every 15-20 days)	750-1000 L	V1	CIB&RC; PHI not stated; TP2
Captan 50% WP / 75% WP / 50% WG	Early blight	2.5 kg, 1667 g or 1500 g per ha (TNAU: captan 0.2%)	750-1000, 1000 or 500 L	V1	CIB&RC; PHI none stated, 8 d, 21 d respectively; TP2
Mancozeb 75% WP; Mancozeb 35% SC	Early blight	1.5-2 kg/ha; 0.5% as printed	750 L; 500 L	V1	CIB&RC; PHI not stated; 35 d
Chlorothalonil 75% WP	Early blight	0.875-1.25 kg/ha	600-800 L	V1	CIB&RC; PHI 14 d
Copper oxychloride 50% WP	Early blight	2.5 kg/ha	750-1000 L	V1	CIB&RC; PHI not stated
Copper sulphate 2.62% SC	Early blight	1 L/ha	500 L	V1	CIB&RC; PHI 3 d
Hexaconazole 2% SC	Early blight	3.0 L/ha	500 L	V1	CIB&RC; PHI 21 d
Kresoxim-methyl 44.3% SC	Early blight	500 ml/ha	500 L	V1	CIB&RC; PHI 25 d
Metiram 70% WG	Early blight	2000 g/ha	500-750 L	V1	CIB&RC; PHI 17 d
Propineb 70% WP	Early blight	0.30% (300 g/100 L)	As required	V1	CIB&RC; PHI 15 d
Ziram 80% WP	Early blight	1.5-2.0 kg/ha	750-1000 L	V1	CIB&RC; PHI 3 d
Kitazin (IBP) 48% EC	Early blight	0.20% (200 ml/100 L)	As required	V1	CIB&RC; PHI 48 d
Aureofungin 46.15% SP	Early blight	0.005% formulation (as printed)	750 L	V1	CIB&RC; PHI 30 d
Azoxystrobin 8.3% + Mancozeb 66.7% WG	Early and late blight	1500 g/ha	500 L	V1	CIB&RC; PHI 25 d
Azoxystrobin 11% + Tebuconazole 18.3% SC	Early and late blight	750 ml/ha	500 L	V1	CIB&RC; PHI not printed
Captan 70% + Hexaconazole 5% WP	Early and late blight	500-1000 g/ha	500 L	V1	CIB&RC; PHI 21 d
Carbendazim 12% + Mancozeb 63% WP	Early blight, late blight, black scurf	1750 g/ha	500 L	V1	CIB&RC; PHI 47 d
Chlorothalonil 12.5% + Mancozeb 62.5% WG	Early and late blight	1750 g/ha	500 L	V1	CIB&RC; PHI 35 d (printed a.i. likely misprint)
Copper oxychloride 47% + Metalaxyl 8% WP	Early and late blight	2127.65 g/ha	700 L	V1	CIB&RC; PHI 34 d
Cyazofamid 6.2% + Mancozeb 60.8% WP	Early and late blight	1250 g/ha	500 L	V1	CIB&RC; PHI 22 d
Kresoxim-methyl 15% + Chlorothalonil 56% WG	Early and late blight	1000 g/ha	500 L	V1	CIB&RC; PHI 23 d
Metalaxyl-M 3.9% + Mancozeb 64% WG	Early and late blight	2500 g/ha	500 L	V1	CIB&RC; PHI 32 d
Propineb 70% + Cymoxanil 6% WP	Early and late blight	2000-2250 g/ha	500 L	V1	CIB&RC; PHI printed as 'at the end of harvest'

Start sprays early, before spots spread, and repeat as advised on the label; TNAU's advice is every 15 to 20 days with zineb or captan (TP2). Rotate chemical groups and keep to the PHI.

Many of the products above are also labelled for potato late blight (marked in the use column), so one programme can cover both (see chapter 1).

6. Sources and citations

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[TP2] TNAU Agritech Portal - Potato: Early blight (Alternaria solani). https://agritech.tnau.ac.in/crop_protection/crop_prot_crop%20diseases_veg_potato_2.html

[AHDB] AHDB (UK) Knowledge Library: Alternaria (early blight) of potato (supplementary, non-Indian source). https://archive.ahdb.org.uk/knowledge-library/alternaria

[SHT] Shtienberg, D. (2013). Alternaria diseases of potatoes: epidemiology and management. Euroblight workshop, Limassol (supplementary, Israel). https://web14.agro.au.dk/project2/euroblight/Workshop/2013Limassol/Proceedings/Page169-180_Shtienberg.pdf

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

3. Black Scurf and Stem Canker

Potato | Rhizoctonia solani (mainly anastomosis group AG 3-PT; sexual stage Thanatephorus cucumeris) | Fungal (seed- and soil-borne; sclerotia)

At a glance. Black, crusty sclerotia on tubers, with reddish-brown cankers girdling sprouts and stems, so poor emergence and misshapen plants. Spread mainly on sclerotia-coated seed tubers. Seed tuber treatment is the registered route and CIB&RC lists several options for potato.

1. Identification

Symptoms

- Tubers: black irregular lumpy encrustations (sclerotia) on progeny tubers, which lower market value (CP1).
- Sprouts and stems: infected sprouts blacken and may fail to emerge; reddish-brown lesions develop on stems and often girdle them (stem canker). Partial or complete girdling causes rosetting of plant tops, purple pigmentation, upward chlorosis or leaf rolling, and aerial tubers can form in leaf axils (CP1).
- The disease can kill eyes during storage and affect stems before emergence (CP1).
- Yield losses up to 35%, mainly through reduced crop stand, are reported (CP1).

Distinguishing features

- Stem canker lesions are reddish-brown and girdle the stem with rosetting and aerial tubers, unlike late blight, which gives white mildew on leaf undersides and stem breakage at nodes (CP1, TP1).

2. Causes and progression

- Pathogen: Rhizoctonia solani Kuhn; AG 3-PT is the group most widespread worldwide on potato, infecting all underground parts at any stage (CP1).
- Survival and spread: both soil- and seed-borne, but the disease reaches new areas mainly on sclerotia-covered seed tubers; soil-borne infection comes later in the season (CP1).
- Progression: the optimum temperature for sclerotial germination is 23 C and for stem lesion development 18 C; maximum sclerotial development on tubers occurs between dehaulming and harvest, and late-harvested crops show more black scurf (CP1).
- Host range is very wide, covering Solanaceae, Fabaceae, Asteraceae, Poaceae and Brassicaceae (CP1).

3. Favourable conditions

- Cool soil (about 18 C) for stem canker and about 23 C for sclerotial germination (CP1).
- Leaving mature tubers in the soil after dehaulming (CP1).
- Short rotations (CP1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Plant clean, certified seed free of sclerotia; sclerotia on seed are the main route into new fields (CP1).
- Rotate: longer rotations lower stem canker and black scurf; rotations with canola or rapeseed lowered Rhizoctonia canker, black scurf and common scab in studies cited by CPRI (CP1).
- Harvest on time after dehaulming (CP1).
- Sulphur fertilisation reduced black scurf and common scab in reports cited by CPRI (CP1).
- Treat seed tubers with Trichoderma viride at 8 g/kg tubers at planting to reduce black scurf and common scab (CP1); a 3% boric acid dip for 30 minutes (or spray) followed by shade drying before cold storage reduces tuber-surface inoculum (CP1).

Regional note

No Maharashtra-specific recommendation was found in the sources reviewed.

5. Remedies and treatment

Biological and non-registered options

- Trichoderma viride at 8 g/kg tubers (ICAR-CPRI); no potato claim exists in the CIB&RC bio-fungicide list (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 23% SC	Black scurf (seed tuber)	1.00 a.i. / 4.00 formulation / 2.5 dilution as printed; units not printed	-	V1	CIB&RC; confirm dose and units on the label
Penflufen 22.43% FS	Black scurf (tuber dip)	0.083 as printed (likely a % solution); 800 kg of seed tubers dipped for 10 minutes, dried in shade, then sown	-	V1	CIB&RC; seed treatment
Pencycuron 22.9% SC	Black scurf (tuber)	0.25-0.50 as printed (units not printed)	-	V1	CIB&RC; PHI 78 d as printed
Thifluzamide 24% SC	Black scurf	2.5 ml/10 kg tubers (0.6 g a.i.)	-	V1	CIB&RC; seed treatment
Carbendazim 25% + Mancozeb 50% WS	Late blight, black scurf (seed)	6-7 g/10 kg seed	2 (units not printed)	V1	CIB&RC; seed treatment
Carboxin 37.5% + Thiram 37.5% WS	Black scurf (seed)	2.5 g/kg seed (1.87 g a.i.)	10 ml/kg slurry	V1	CIB&RC; seed treatment, no waiting period
Sedaxane 3.70% + Fludioxonil 4.63% FS	Black scurf (seed tuber)	0.40-0.50 ml/kg seed	2-3 L water per acre of tubers as printed	V1	CIB&RC; seed treatment
Tebuconazole 15% + Zineb 57% WDG	Black scurf (seed dresser)	20 g/100 kg seed tubers	Enough to coat	V1	CIB&RC; seed dresser
Thiophanate-methyl 44.8% + Kasugamycin 26% SC	Black scurf (seed)	3.5 ml/10 kg tuber seed	200 ml/10 kg tubers as printed	V1	CIB&RC; seed treatment
Carbendazim 12% + Mancozeb 63% WP	Black scurf with early and late blight (foliar)	1750 g/ha	500 L	V1	CIB&RC; PHI 47 d
Trichoderma viride (formulation)	Black scurf and common scab	8 g/kg tubers at planting	-	V2	ICAR-CPRI (CP1); not a potato label claim in CIBB
Boric acid	Tuber-surface inoculum	3% dip 30 minutes or spray; shade dry before cold storage	-	V2	ICAR-CPRI (CP1); not on the CIB&RC list

Dose units are not printed for some CIB&RC rows (marked as printed); follow the label.

Seed tuber treatments have no PHI because they are applied before planting.

6. Sources and citations

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[TP1] TNAU Agritech Portal - Potato: Late blight (Phytophthora infestans). https://agritech.tnau.ac.in/crop_protection/crop_prot_crop%20diseases_veg_potato_1.html

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

4. Common Scab

Potato | Streptomyces scabies and other Streptomyces spp. (more than 19 reported) | Bacterial (actinomycete; seed- and soil-borne)

At a glance. Rough, corky, sometimes pitted lesions on the tuber skin that seldom cut yield but lower price, and seed lots above 5% incidence are rejected by Indian certification. Infection happens in the weeks after tuber initiation, so soil moisture and pH management matter most.

1. Identification

Symptoms

- Begins as small reddish or brownish spots on the tuber surface; infection takes place through lenticels, and the surrounding periderm turns brown and rough, then corky (CP1).
- Indian symptom types: (1) brownish roughening or abrasion of the skin; (2) proliferated lenticels with hard corky deposits, sometimes star-shaped; (3) raised, rough, corky pustules; (4) pits 3 to 4 mm deep ringed by hard corky tissue; (5) concentric wrinkled layers around a black core (CP1).
- The disease seldom affects yield, but affected tubers fetch low prices; seed lots above 5% incidence are rejected by Indian seed certification (CP1).

Distinguishing features

- Black scurf gives irregular lumpy black crusts (sclerotia) rather than corky pitted lesions (CP1).
- Silver scurf shows light brown or greyish leathery spots, mostly at the stem end (CP1).

2. Causes and progression

- Pathogen: pathogenic Streptomyces species produce thaxtomin phytotoxins that cause hypertrophy and cell death; Streptomyces scabies, S. acidiscabies, S. turgidiscabies and others are named (CP1).
- Infection: potato is most susceptible in the period after tuber initiation, through stomata and immature lenticels; once the periderm has differentiated, tubers are no longer susceptible (CP1).
- Survival: seed- and soil-borne; the organism persists in soil for years on decaying organic matter and manure, and infected tubers are the field inoculum source (CP1).

3. Favourable conditions

- Soil pH from 5.2 to 8.0 or more, temperature of 20 to 30 C and low soil moisture; the pathogen is aerobic (CP1).
- Soil pH near neutral is linked to higher scab (CP1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Keep soil moist for 10 to 20 days after tuber initiation, because high soil moisture then reduces common scab (CP1).
- Lower the soil pH or use manganese; sulphur fertilisation also reduced scab and black scurf (CP1).
- Rotation: the pathogen population fell when the field was left fallow after potato or when mung bean or sunn hemp were grown as green manure over four years; canola and rapeseed rotations also reduced scab (CP1).
- Treat seed tubers with Trichoderma viride at 8 g/kg tubers, or with Bacillus subtilis, both reported to reduce scab severity (CP1).
- Use certified seed lots below the 5% threshold (CP1).

Regional note

No Maharashtra-specific recommendation was found in the sources reviewed.

5. Remedies and treatment

Biological and non-registered options

- Trichoderma viride (8 g/kg tubers) and Bacillus subtilis tuber treatment are ICAR-CPRI reports (CP1); the CIB&RC bio-fungicide list has no potato entry (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Thiram 75% WS	Potato scab (seed)	25-30 g formulation (18.8-22.5 g a.i.); seed quantity unit not legible in the list	1 as printed	V1	CIB&RC; PHI printed as 7-10
Trichoderma viride; Bacillus subtilis (tuber treatment)	Common scab	T. viride 8 g/kg tubers at planting	-	V2	ICAR-CPRI (CP1); not potato label claims in CIBB

Thiram is the only CIB&RC fungicide row for potato scab. Follow the label for the seed quantity the dose applies to.

Management is mostly cultural; fungicide seed treatment does not replace soil moisture and rotation management.

6. Sources and citations

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

5. Bacterial Wilt (Brown Rot)

Potato | Ralstonia solanacearum (Phylotype IIB sequevar 1, formerly race 3 / biovar 2) | Bacterial (soil- and tuber-borne)

At a glance. Sudden unrecoverable wilting of plants on hot days, with brown vascular rings and sticky bacterial ooze in tubers. Latently infected seed tubers spread it to new areas. Prevention through clean seed, clean soil and rotation is the base; streptocycline and stable bleaching powder have defined roles.

1. Identification

Symptoms

- Earliest sign: slight wilting of leaves of the top branches on hot sunny days; leaves droop and then wilt completely and irreversibly (CP1).
- Field test: place a cut stem section in clear water; within minutes whitish thread-like bacterial streaming comes from the cut end (CP1).
- Tubers: a transverse cut shows water-soaked brown vascular circles with dirty-white sticky drops in about 2 to 3 minutes; in advanced stages bacterial mass oozes from the eyes (CP1).
- The disease also causes rot of tubers in field, store and transit (CP1).

Distinguishing features

- Fungal wilts do not give bacterial streaming in clear water (CP1).
- Soft rot gives a slimy, foul-smelling tuber rot and blackleg gives stem necrosis from the seed tuber upward (CP1).

2. Causes and progression

- Pathogen: the Gram-negative bacterium Ralstonia solanacearum; the strain usually linked to potato is R. solanacearum Phylotype IIB sequevar 1 (CP1).
- Entry: the bacterium infects roots of healthy plants through wounds; infected tubers and plant debris in infested soil are the two main inoculum sources (CP1).
- Latent infection: plants may carry the bacterium without symptoms, so latently infected seed tubers can spread it from seed production sites to healthy potato areas (CP1).
- Host range: more than 200 species in over 50 plant families (CP1).

3. Favourable conditions

- Mean soil temperature below 15 C or above 35 C does not favour disease (CP1).
- Soil moisture raises survival, infection, disease development and spread (CP1).
- Mostly reported in acidic soils (pH 4.3 to 6.8), and occasionally in alkaline soils (CP1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use healthy, certified seed tubers; latent infection makes seed source critical (CP1).
- Avoid fields with a history of the disease; short rotations raise potato diseases in general (CP1).
- Clean the field of weeds and remnants and deep-plough in summer or winter (CP1).
- Apply stable bleaching powder at 12 kg/ha in the furrow along with fertiliser at planting (CP1).

Regional note

No Maharashtra-specific recommendation was found in the sources reviewed.

5. Remedies and treatment

Biological and non-registered options

- Pseudomonas fluorescens 1.0% WP (IIHR-PF-2) is on the CIB&RC bio-fungicide list for bacterial wilt of tomato and brinjal, not potato (CIBB); it is not recommended for potato here.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Streptomycin sulphate 90% + tetracycline HCl 10% SP (Streptocycline); also the 9% + 1% SP	Blackleg and soft rot, bacterial brown wilt or ring (bangle) disease of potato	Seed tubers: soak 30 minutes in 40-100 ppm before planting. Spray: 2-3 sprays of 40-50 ppm at 20-day intervals, first spray 30 days after planting	-	V1	CIB&RC (potato entry); PHI not printed
Stable bleaching powder	Bacterial wilt and soft rot	12 kg/ha in the furrow at planting, with fertiliser	-	V2	ICAR-CPRI (CP1); not on the CIB&RC list

Streptocycline is labelled on potato for this disease group; use it as a seed soak plus sprays exactly as the label states.

Clean seed and clean fields remain the main defence; chemicals do not clear the bacterium from infested soil.

6. Sources and citations

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

6. Soft Rot and Blackleg

Potato | Pectobacterium atrosepticum, P. carotovorum, P. brasiliense, Dickeya spp. (syn. Erwinia) | Bacterial (tuber- and soil-associated)

At a glance. Slimy, foul-smelling tuber rot after wounds or poor storage, and blackening of the stem base (blackleg) on growing plants. Losses can reach 100% in poorly ventilated stores and transit. Avoid injury, keep tubers dry and ventilated, and treat seed carefully.

1. Identification

Symptoms

- Blackleg: necrosis of the stem starting where it meets the mother tuber and spreading several centimetres above ground; the vascular system becomes necrotic, leaves wilt and curl, and the stem pith may rot (CP1).
- Tubers: small soft, water-soaked spots around lenticels that enlarge under high humidity, sometimes with a brown to black margin; rotten tubers become slimy and foul-smelling, brown liquid oozes, the skin often stays intact and tubers may swell with gas (CP1).
- At harvest many small rotten tubers with intact skin may be seen; the disease affects the crop at all stages but is most serious on tubers in poor storage in warm, wet climates (CP1).

Distinguishing features

- Bacterial wilt shows bacterial streaming in clear water and brown vascular rings (CP1).
- Fusarium dry rot gives dry, shrunken, wrinkled tuber rot, not slimy rot (CP1).

2. Causes and progression

- Pathogens: Pectobacterium atrosepticum (syn. Erwinia carotovora subsp. atroseptica), P. carotovorum, P. brasiliense and Dickeya species; temperature limits differ (P. atrosepticum and P. parmentieri cannot survive above 33 C, while P. carotovorum and P. brasiliense grow up to 39 C) (CP1).
- Entry: soft rot bacteria infect tubers damaged by mechanical injury or by other tuber-borne pathogens; blackleg occurs when bacteria enter stems harmed by tools or weather (CP1).
- Spread: bacteria can be carried latently in lenticels, wounds and on the tuber surface, then spread to healthy tubers in stores and during seed cutting, handling and planting; decaying tubers in soil are a source (CP1). Occurrence on tubers tends to increase with each generation of multiplication (CP1).
- Losses: up to 100% in poorly handled or poorly ventilated produce in store or transit (CP1).

3. Favourable conditions

- Water film on tuber surfaces that proliferates lenticels and creates anaerobic conditions, plus injury (CP1).
- Tubers harvested in wet soil, poor ventilation in transit and store, and warm, wet climates (CP1).
- Short rotations in warm climates and poorly drained soil (CP1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Avoid injury at harvest, handling and seed cutting; do not cut seed tubers (CP1).
- Keep tubers dry and well ventilated in store and transit (CP1).
- Use clean, certified seed and avoid dumping rotten tubers near fields (CP1).
- Seed tuber treatment with 3% boric acid (30 minutes dip or spray), followed by shade drying before cold storage, reduces tuber-surface fungal and bacterial inoculum (CP1).
- Apply stable bleaching powder at 12 kg/ha in the furrow at planting (CP1).

Regional note

No Maharashtra-specific recommendation was found in the sources reviewed.

5. Remedies and treatment

Biological and non-registered options

- No potato bio-fungicide or bactericide claim exists in the CIB&RC bio list (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Streptomycin sulphate 90% + tetracycline HCl 10% SP (Streptocycline); also the 9% + 1% SP	Blackleg and soft rot of potato	Seed tubers: soak 30 minutes in 40-100 ppm before planting. Spray: 2-3 sprays of 40-50 ppm at 20-day intervals, first spray 30 days after planting	-	V1	CIB&RC (potato entry); PHI not printed
Stable bleaching powder	Bacterial wilt and soft rot	12 kg/ha in furrow at planting	-	V2	ICAR-CPRI (CP1)
Boric acid (seed tuber dip)	Tuber-surface bacterial and fungal inoculum	3% dip for 30 minutes or spray; shade dry before cold storage	-	V2	ICAR-CPRI (CP1)

Prevention by handling and storage conditions matters more than chemicals for this disease.

6. Sources and citations

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

7. Fusarium Dry Rot

Potato | Fusarium spp. (15 species reported; F. sambucinum, F. solani and others) | Fungal (soil- and debris-borne; tuber storage rot)

At a glance. A storage and seed-piece rot that shrinks tubers into dry, wrinkled, sunken masses with powdery growth, entering through wounds. Rotation helps little because Fusarium has a wide host range. Minimising wounds and good storage are the main controls; no fungicide row for potato dry rot is on the CIB&RC list.

1. Identification

Symptoms

- Affected tubers dry out, shrink and wither, with a paper-thin light to dark brown skin; the diseased area is soft and spongy, with a bad odour (CP1).
- A black, dry, powdery fungal growth appears on the tuber surface; a longitudinal cut shows internal rot, necrotic lesions and tan-brown to white or pinkish mycelium (CP1).

Distinguishing features

- Soft rot is slimy and wet, whereas dry rot gives dry, wrinkled tissue (CP1).
- Silver scurf is a skin blemish without internal rot (CP1).

2. Causes and progression

- Pathogens: fifteen Fusarium species have been identified as dry rot agents (CP1).
- Survival: chlamydospores and other propagules persist in soil and plant debris and are the main inoculum source for field tuber infection; progeny tubers rot after wounds from intercultural operations (CP1).
- Temperature: mycelial growth is best at 20 to 25 C and sporulation at 25 to 30 C; temperatures below 5 C generally reduce risk (though F. sambucinum and F. graminearum still grow); above 40 C all species are inhibited (CP1).

3. Favourable conditions

- Wounds from handling and field operations (CP1).
- 20 to 30 C for growth and sporulation (CP1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Minimise wounds at harvest, grading and cutting; avoid cutting seed tubers (CP1).
- Crop rotation has limited effect because Fusarium infects many hosts and persists in soil (CP1).
- Adequate potassium reduces incidence of dry rot (CP1).
- Seed tuber treatment with 3% boric acid (30-minute dip or spray) before cold storage reduces tuber-surface inoculum (CP1).

Regional note

No Maharashtra-specific recommendation was found in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Boric acid (seed tuber dip)	Tuber-surface inoculum incl. dry rot	3% dip 30 minutes or spray; shade dry before storage	-	V2	ICAR-CPRI (CP1); not on the CIB&RC list
Foliar or soil fungicides	Fusarium dry rot (chemical)	-	-	--	No potato claim for this disease in the CIB&RC list and no verified recommendation; not recommended

No CIB&RC fungicide row for potato dry rot was found. Rely on handling, storage conditions and the boric acid dip described by ICAR-CPRI.

6. Sources and citations

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

8. Silver Scurf

Potato | Helminthosporium solani | Fungal (tuber-borne skin blemish; spreads in storage)

At a glance. A skin blemish disease: light brown to grey, lustrous leathery spots, mostly at the stem end, that spread in humid stores and reduce market value. Infected seed tubers can show severe disease within a week after going into store.

1. Identification

Symptoms

- Light brown or greyish leathery spots, mainly at the stem end of tubers, that can expand over the whole tuber; the lustrous appearance is more noticeable when tubers are moist, but also visible when dry (CP1).

Distinguishing features

- Common scab gives rough, corky, pitted lesions (CP1); silver scurf spots are light brown or greyish and leathery, mostly at the stem end (CP1).

2. Causes and progression

- Pathogen: Helminthosporium solani; a skin-blemish disease that has emerged as significant over the past two decades, in the field and in storage (CP1).
- Spread: spores produced below the soil surface are carried down by irrigation and rain to stolons, roots and progeny tubers, where abundant sporulation recurs within a week, giving multiple infection cycles in soil (CP1).
- Storage: infected seed tubers may show few or no symptoms at harvest but develop severe infection within a week after moving into warehouses or cold stores; repeated sporulation cycles occur at 85 to 95% relative humidity (CP1).

3. Favourable conditions

- Infection needs at least about 3 C and 90% relative humidity; high humidity (85 to 95%) in storage drives repeated cycles (CP1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use clean seed; keep stores from becoming humid and warm (CP1).
- Seed tuber treatment with 3% boric acid (30-minute dip or spray) followed by shade drying reduces tuber-surface inoculum (CP1).

Regional note

No Maharashtra-specific recommendation was found in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Boric acid (seed tuber dip)	Tuber-surface inoculum	3% dip 30 minutes or spray; shade dry before cold storage	-	V2	ICAR-CPRI (CP1); not on the CIB&RC list
Foliar or soil fungicides	Silver scurf (chemical)	-	-	--	No potato claim for this disease in the CIB&RC list and no verified recommendation; not recommended

No CIB&RC fungicide row for silver scurf was found.

6. Sources and citations

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used:

https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

9. Potato Viruses (Leaf Roll, Y, X and Apical Leaf Curl)

Potato | Potato leaf roll virus (PLRV), Potato virus Y (PVY), Potato virus X (PVX), potato apical leaf curl virus | Viral (aphid-transmitted; tuber-borne in seed)

At a glance. Viruses build up in seed tubers from season to season, reducing yield and seed quality. PLRV and PVY are the most important aphid-transmitted viruses for seed production in India. There is no cure; healthy seed, seed plot technique, aphid management and rogueing are the measures. Fungicides have no role.

1. Identification

Symptoms

- Leaf roll (PLRV): typical leaf rolling with chlorosis and stunting; the disease is widespread in India and especially serious in the eastern plain zone (VIR2).
- PVY: symptoms vary with strain and variety, from mild mottling to vein necrosis, leaf drop, mosaic and crinkling (VIR4, AGR).
- PVX: mild or interveinal mosaic, often latent in many varieties; PVX together with PVY causes more harm than either alone (AGR, VIR1).
- Potato apical leaf curl virus and severe or mild mosaic were scored in AICRP Hisar trials (VIR1).
- Viruses can infect potato without obvious symptoms, and yield reductions of 20 to 80% are reported depending on virus and variety (AGR).

Distinguishing features

- Early blight and late blight give necrotic lesions on leaves, not systemic rolling or mosaic (CP1).
- Leaf-roll type symptoms can also come from tuber-borne infection, which is why the tuber source matters (VIR2).

2. Causes and progression

- PLRV: a Polerovirus (Luteoviridae), spread by aphids; the green peach aphid transmits it persistently; global losses up to 90% are cited (VIR2, VIR4, AGR).
- PVY: transmitted non-persistently by aphids, in seconds; it interacts with PVX (VIR4, AGR).
- Vectors in India: fourteen aphid species were found on potato across all zones; Myzus persicae and Aphis gossypii were the most abundant and widespread, and species other than M. persicae can matter for virus incidence in seed crops (VIR3). Foxglove aphid (Aulacorthum solani) also carries PVY and PLRV (supplementary record).
- Seed: the viruses build up quickly in potato stocks through tubers (VIR4); degeneration is assessed in AICRP seed trials (VIR1).

3. Favourable conditions

- In the eastern plain zone, aphid populations are very high in mid-February and March; higher temperatures later reduce aphids and break the leaf roll cycle (VIR2).
- Aphid vectors and the use of seed from earlier generations raise virus incidence (VIR1, VIR3).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Plant healthy, virus-free certified seed, ideally from tissue culture and meristem culture, produced at high altitude or in areas with low aphid populations (VIR2, AGR).
- Seed plot technique (SPT) seed had lower leaf roll incidence than previous-year seed in AICRP Hisar trials (VIR1).
- Manage aphid vectors, and rogue virus-infected plants (VIR2, AGR).
- Use virus-resistant genotypes where available; seven of 100 germplasm lines were resistant to potato apical leaf curl New Delhi virus in AICRP trials (VIR1).

Regional note

No Maharashtra-specific recommendation was found. The eastern plain zone note applies to that region (VIR2).

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Fungicides (any)	Viral diseases	Not applicable	-	--	Fungicides do not control viruses; no CIB&RC fungicide row applies
Insecticides for aphid vectors	PVY, PLRV vector management	Not verified here	-	--	Registration of aphid insecticides on potato not checked in this document; seek label advice

Virus control rests on seed health and vector management. ICAR-CPRI and AICRP Potato centres supply guidance on seed plot technique and aphid thresholds.

6. Sources and citations

[VIR1] Selection of potato apical leaf curl virus disease-resistant genotypes, pest surveillance, seed degeneration and management of aphid in field conditions (AICRP on Potato, Hisar). Innovare Academic Sciences. https://journals.innovareacademics.in/index.php/ijags/article/download/44223/26226

[VIR2] First report on molecular basis of potato leaf roll virus (PLRV) aggravation by combined effect of tuber and prevailing aphid (India; PMC7657319). https://pmc.ncbi.nlm.nih.gov/articles/PMC7657319

[VIR3] Sridhar, J. et al. (2022). Species composition and distribution of the vector aphids of PVY and PLRV in India. Potato Research 65(3): 601-617 (ICAR-CPRI; AGRIS record). https://agris.fao.org/search/fr/records/65df2f0f63b8185d9cab7141

[VIR4] University of Maine Cooperative Extension Bulletin #2492: Potato Facts - Potato diseases caused by PVY and PLRV (supplementary, non-Indian). https://extension.umaine.edu/publications/2492e/

[AGR] Agridots: Diseases of potato - late blight, wart, viruses (educational compilation; supplementary, non-primary source). https://agridots.com/courses/bsc-agriculture/plant-pathology/path373-diseases-of-field-and-hort-crops-management-ii/09-diseases-of-potato

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

10. Potato Wart (Quarantine Disease)

Potato | Synchytrium endobioticum | Fungal (soil-borne obligate biotroph; long-lived resting spores)

At a glance. Cauliflower-like warty galls on tubers, stolons and other meristematic tissues. In India it is a notifiable quarantine disease restricted to the Darjeeling hills of West Bengal. This chapter rests on international and supplementary sources and carries no chemical recommendation.

1. Identification

Symptoms

- Typical symptoms are cauliflower-like galls that can form on all meristematic tissues of potato except roots (SPR).
- Above-ground symptoms are rare, and the disease is usually detected at harvest (AGR).

Distinguishing features

- Galls on tubers are warty outgrowths, unlike the corky pits of common scab (CP1, SPR).

2. Causes and progression

- Pathogen: Synchytrium endobioticum, a soil-borne biotrophic fungus that causes potato wart (SPR).
- Quarantine: wart is a notifiable disease in India, restricted to the Darjeeling hills of West Bengal (AGR; supplementary source).

3. Favourable conditions

- The sources reviewed give no weather thresholds.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Follow quarantine rules: do not move tubers or soil out of affected areas, use certified seed from approved sources, and report suspected wart to plant protection authorities (AGR, quarantine status).

Regional note

Not reported from Maharashtra in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Foliar or soil fungicides	Potato wart	-	-	--	No potato claim for this disease in the CIB&RC list and no verified recommendation; not recommended

No chemical recommendation is given; control is through quarantine and clean seed.

6. Sources and citations

[SPR] Major potato diseases: late blight, early blight, wart and powdery scab (book chapter abstract, Springer; supplementary, international). https://link.springer.com/chapter/10.1007/978-3-030-28683-5_9

[AGR] Agridots: Diseases of potato - late blight, wart, viruses (educational compilation; supplementary, non-primary source). https://agridots.com/courses/bsc-agriculture/plant-pathology/path373-diseases-of-field-and-hort-crops-management-ii/09-diseases-of-potato

[CP1] Sharma, S., Sagar, V. & Lal, M. (2024). Disease management in seed potato production. Indian Farming 74(5): 52-58 (ICAR-Central Potato Research Institute, Shimla and Modipuram). https://epubs.icar.org.in/index.php/IndFarm/article/download/151964/54634/414099

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

Appendix A. Cross-check log

- Potato rows in the CIB&RC fungicide list (up to 30.09.2025) were read in full: single products and combinations for late blight and early blight, seed tuber treatments for black scurf, thiram for scab, and streptocycline for blackleg, soft rot and brown rot / ring disease.
- Items printed in a way that needs care: chlorothalonil 12.5% + mancozeb 62.5% WG lists chlorothalonil a.i. as 2178.75 g (likely 218.75 g); PHIs printed as 'at the end of harvest' (propineb + cymoxanil), '3-5' (mancozeb 75 WG) and 'not less than 7 weeks' (metalaxyl 8% + mancozeb 64%); pencycuron lists 78 days and units that were not printed; azoxystrobin 23% SC black scurf units not printed; famoxadone + cymoxanil and oxathiapiprolin + amisulbrom formulation doses are printed without units.
- The CIB&RC bio-fungicide list (up to 30.09.2025) has no entry for potato; its tomato and brinjal entries are not reproduced for potato.
- No CIB&RC row exists for potato Fusarium dry rot, silver scurf, viruses or wart. Fungicides do not control viruses.
- Model classes (early blight, late blight, healthy) are covered by chapters 2 and 1 and Appendix B.

Appendix B. Image-classification guide for the model classes

The table maps the potato classes used by an image model (early blight, late blight) and a healthy-leaf class to the chapters above, with visual features and the look-alikes most likely to cause confusion. Confirm field diagnoses by looking for white sporulation under the leaf, the pattern of the lesion margin and leaves of different ages, because lesions can look alike in one photo.

Class	Chapter	What it looks like	Most likely confusion	How to separate	Management pointer
Early blight	2	Dry, papery, small round, oval or angular dark brown to black spots limited by veins, often with concentric rings; yellowing round the spots; spots can coalesce (CP1, TP2).	Late blight; magnesium deficiency, manganese toxicity, Verticillium wilt	No water-soaked margin and no white sporulation on the leaf underside; spores cannot be seen even with a hand lens (AHDB).	Chapter 2 table: zineb, captan, mancozeb and other V1 products
Late blight	1	Water-soaked, pale green irregular lesions near leaf tips and margins turning brown to purplish-black; white mildew under the leaf edge in humid weather; brown stem lesions (CP1, TP1).	Early blight	Water-soaked, spreading lesions with white sporulation beneath; rapid collapse in cool humid weather (CP1).	Chapter 1 table: many V1 options; start protection before disease
Healthy leaf	-	No necrotic lesions, no white mildew, no rolling or mosaic. The sources in this document do not describe healthy potato foliage.	Early disease, natural ageing of lower leaves, nutrient problems	General caution, not from a cited source: a few yellowing lower leaves can be natural ageing; mosaic or rolling without lesions points to viruses (chapter 9).	Nothing to treat; keep scouting